

Algebraic Geometry: Scheme-based

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1 Introduction

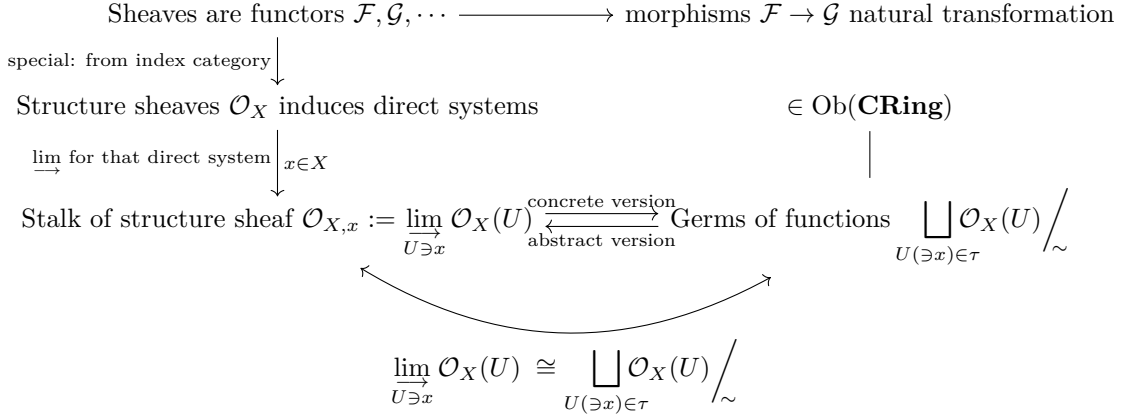
2 Sheaves and presheaves

2.1 Basic definitions

Definition 2.1: Induced sheaf

Let $(X, \tau, \mathcal{O}_X)$ be a locally ringed space and $U \overset{\text{open}}{\subseteq} X$. The **induced sheaf** by U , denoted $\mathcal{O}_X|_U$, is defined elementwise:

$$\forall V \overset{\text{open}}{\subseteq} U, \quad \mathcal{O}_X|_U(V) := \mathcal{O}_X(V)$$



Definition 2.2: Pushforward sheaf

Let $(X, \tau, \mathcal{O}_X)$ and $(Y, \sigma, \mathcal{O}_Y)$ be two ringed spaces. Let $f : X \rightarrow Y$ be a continuous map. The **pushforward** or **direct image** of $\mathcal{O}_X$, denoted $f_*\mathcal{O}_X$, is defined elementwise as

$$\forall U \overset{\text{open}}{\subseteq} Y, \quad f_*\mathcal{O}_X(U) := \mathcal{O}_X(f^{-1}(U))$$

$f_*\mathcal{O}_X$ is also called the **pushforward** of $\mathcal{O}_X$ by f .

Lemma 2.0.1 *This so-called pushforward sheaf $f_*\mathcal{O}_X$ is a sheaf.*

Proof: (1) $\forall U \overset{\text{open}}{\subseteq} Y$, since f is continuous, $f^{-1}(U) \overset{\text{open}}{\subseteq} X$. Then by definition,

$$f_*\mathcal{O}_X(U) = \mathcal{O}_X(f^{-1}(U))$$

is a set(abelian group, commutative ring), because $\mathcal{O}_X$ is a sheaf: $\forall V \overset{\text{open}}{\subseteq} X$, $\mathcal{O}_X(V)$ is a set(abelian group, commutative ring).

(2) $\forall U \overset{\text{open}}{\subseteq} Y, \forall V \subseteq U$, with V open in X , $\exists$ a 'function' ^[1] $\text{res}_{U,V} : f_*\mathcal{O}_X(U) \rightarrow f_*\mathcal{O}_X(V)$, a restriction morphism, has two additional properties: $\bullet \forall U \overset{\text{open}}{\subseteq} X, \text{res}_{U,U} = \text{id}_{f_*\mathcal{O}_X(U)}$

$\bullet$ If $W \subseteq V \subseteq U$, with $W, V, U \overset{\text{open}}{\subseteq} X$, then $\text{res}_{U,W} = \text{res}_{V,W} \circ \text{res}_{U,V}$

It is not hard to verify that $\bullet$ the existence of $\text{res}_{U,V}$ and $\bullet$ those two properties come from definition below:

$$\text{res}_{U,V} := \text{res}_{f^{-1}(U), f^{-1}(V)}$$

(3) (Locality) $f^{-1}(U) \overset{\text{open}}{\subseteq} X$ and let $\{U_i\}_{i \in I}$ be any open cover of U , with $\forall i, U_i \subseteq U$. Then $\{f^{-1}(U_i)\}_{i \in I}$ is an open cover of $f^{-1}(U)$. To be compatible with the notation before, $\forall s, t \in f_*\mathcal{O}_X(U) = \mathcal{O}_X(f^{-1}(U))$:

$$\text{res}_{U, U_i}(s) = \text{res}_{U, U_i}(t) \quad \text{assumption} \quad (1)$$

$$\Rightarrow \text{res}'_{f^{-1}(U), f^{-1}(U_i)}(s) =: \text{res}_{U, U_i}(s) = \text{res}_{U, U_i}(t) := \text{res}'_{f^{-1}(U), f^{-1}(U_i)}(t) \quad \text{definition} \quad (2)$$

$$\Rightarrow \text{res}'_{f^{-1}(U), f^{-1}(U_i)}(s) = \text{res}'_{f^{-1}(U), f^{-1}(U_i)}(t) \quad (3)$$

$$\Rightarrow s = t \quad \text{locality of } \mathcal{O}_X \quad (4)$$

(4) (Gluing) $f^{-1}(U) \stackrel{\text{open}}{\subseteq} X$ Let $\{U_i\}_{i \in I}$ be any open cover of U , with $\forall i, U_i \subseteq U$. Then $\{f^{-1}(U_i)\}_{i \in I}$ is an open cover of $f^{-1}(U)$. $\{s_i \in \mathcal{O}_X(U_i)\}_{i \in I}$ is a family of sections.

$$\forall i, j \in I, s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \Rightarrow \exists s \in \mathcal{O}_X(U) : \forall i \in I, s|_{U_i} = s_i$$

$$\text{res}_{U, U_i \cap U_j}(s_i) = \text{res}_{U, U_i \cap U_j}(s_j) \quad \text{assumption} \quad (5)$$

$$\Rightarrow \text{res}'_{f^{-1}(U), f^{-1}(U_i \cap U_j)}(s_i) =: \text{res}_{U, U_i}(s_i) = \text{res}_{U, U_i}(s_j) := \text{res}'_{f^{-1}(U), f^{-1}(U_i \cap U_j)}(s_j) \quad \text{definition} \quad (6)$$

$$\Rightarrow \text{res}'_{f^{-1}(U), f^{-1}(U_i) \cap f^{-1}(U_j)}(s_i) = \text{res}'_{f^{-1}(U), f^{-1}(U_i) \cap f^{-1}(U_j)}(s_j) \quad (7)$$

$$\Rightarrow \exists s \in \mathcal{O}_X(f^{-1}(U)) : \forall i \in I, \text{res}'_{f^{-1}(U), f^{-1}(U_i)} = s_i \quad \text{gluability in } \mathcal{O}_X \quad (8)$$

$$\Leftrightarrow \exists s \in f_* \mathcal{O}_X(U) : \forall i \in I, \text{res}_{U, U_i} = s_i \quad \text{definition} \quad (9)$$

□

2.2 Local information: Stalk and germs

Definition 2.3: Stalk of a sheaf

Let $(X, \tau, \mathcal{O}_X)$ be a ringed space. Given $x \in X$, the **stalk** of the sheaf $\mathcal{O}_X$ **at** x , denoted $\mathcal{O}_{X,x}$, is defined as

$$\mathcal{O}_{X,x} := \varinjlim_{U \ni x} \mathcal{O}_X(U)$$

Remark (1) To see that this definition is well-defined, we have **Sheaf induces a direct system**. Then it is possible to define the notion of direct limit.

(2) This is a direct limit in the category **CRing**.

(3) In general, $\mathcal{O}_{X,x}$ is merely a commutative ring. We do not know whether it is a local ring up to now.

Stalk **generalises** the notion of germs of functions.

Let $x \in X$ and U be an open neighbourhood of x . The germs of functions of $f \in \mathcal{O}_X(U)$ at x is defined as the followings:

For $f \in \mathcal{O}_X(V)$ and $g \in \mathcal{O}_X(W)$, with $V, W \subseteq U$, $(f, V) \sim (g, W) \Leftrightarrow V, W$ are open neighbourhood of x and $f|_{V \cap W} = g|_{V \cap W}$. Then, $[(f, U)]$ is defined to be the equivalence class of (f, U) . Sometimes, we use another notation $f_x := [(f, U)]$. Let $\text{Germ}_x(X)$ be the collection of all germs of functions at x . Explicitly,

$$\text{Germ}_x(X) = \{f_x | x \in X\} = \{[(f, U)] | f \in \mathcal{O}_X(U), U \text{ is an open neighbourhood of } x\}$$

- The germs of functions at x is a stalk because it is a direct limit of the direct system $\{\mathcal{O}_X(U)\}_U$:

$$\begin{array}{ccc} \mathcal{O}_X(U) & \xrightarrow{\text{res}_{U,V}} & \mathcal{O}_X(V) \\ \searrow \varphi_U & & \swarrow \varphi_V \\ & \text{Germ}_x(X) & \\ \swarrow \psi_U & \downarrow \exists! \Phi & \searrow \psi_V \\ & Y & \end{array}$$

Define $\varphi_U : f \mapsto [(f, U)]$ for all open neighbourhood of x , U . Clearly, $\varphi_U = \varphi_V \circ \text{res}_{U,V}$. Also, for any commutative ring Y , suppose that ψ_U is another ring homomorphism such that $\psi_U = \psi_V \circ \text{res}_{U,V}$. Then, define $\Phi : [(f, U)] \mapsto \psi_U(f)$. This is well-defined because for another representative $[(g, V)] = [(f, U)]$ such that $V \subseteq U$, $\psi_U(f) = (\psi_V \circ \text{res}_{U,V})(f) = \psi_V(g)$ by $f|_V = g|_V$. This Φ is also a ring homomorphism because ψ_U is. The definition of Φ automatically makes the diagram being commutative. Moreover, Φ is unique. ??

Importance of stalk and relations between sheaf and stalk:

- Stalk $\Rightarrow$ Sheaf: In many situations, knowing the stalks of a sheaf is enough to control the sheaf itself.
- Sheaf $\Rightarrow$ Stalk: the global information present in a sheaf typically carry less information.

Definition 2.4: Locally-ringed space

A ringed space $(X, \tau, \mathcal{O}_X)$ is called a locally ringed space $\Leftrightarrow \forall x \in X, \mathcal{O}_{X,x}$ is a local ring.
In the language of stalk 2.3, a locally ringed space is a ringed space $(X, \tau, \mathcal{O}_X)$ such that all stalks of $\mathcal{O}_X$ are local rings.

Elements of $(X, \mathcal{O}_X)$ are denoted as $(U, f) \in (X, \mathcal{O}_X)$.

Example (1)

(2) A Riemann surface $\widehat{\mathbb{C}}$ with its sheaf $\mathcal{O}_{\widehat{\mathbb{C}}}$ of analytic functions, i.e. $(\widehat{\mathbb{C}}, \mathcal{O}_{\widehat{\mathbb{C}}})$

Given a morphism of ringed spaces, $(f, f^\#) : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$.

We have a collection of morphisms: $f_U^\# : \mathcal{O}_Y(U) \rightarrow f_*\mathcal{O}_X(U)$. Consider the direct limit $\mathcal{O}_{Y,f(x)}$ and $\mathcal{O}_{X,x}$, we want $f^\#$ somehow connects the two local rings, in other words, ring homomorphism between them.

Let $I := \{U \subseteq X | x \in U \text{ open sets}\}$ and $J := \{V \subseteq Y | f(x) \in V \text{ open sets}\}$.

Since $f^\#$ gives a morphism between two collections of rings, it naturally induces a map

$$\mathcal{O}_{Y,f(x)} := \varinjlim_{V \in J} \mathcal{O}_Y(V) \longrightarrow \varinjlim_{V \in J} f_*\mathcal{O}_X(V) = \varinjlim_{V \in J} \mathcal{O}_X(f^{-1}(V)) \xrightarrow{\dagger} \varinjlim_{U \in I} \mathcal{O}_X(U) =: \mathcal{O}_{X,x}$$

$\dagger$: This is because $\{f^{-1}(V)\}_{V \in J} \subseteq \{U\}_{U \in I} = I$, since f is continuous

$\subseteq$: $\forall V \in J, f^{-1}(V)$ is open in X and $f(x) \in V \Rightarrow x \in f^{-1}(V)$ so $f^{-1}(V) \in I$

Definition 2.5: Morphism between locally ringed spaces

Let $(X, \tau, \mathcal{O}_X)$ and $(Y, \sigma, \mathcal{O}_Y)$ be two locally ringed spaces. A morphism from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$, denoted

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

given by the following data:

- A continuous map $f : X \rightarrow Y$.
- A morphism of sheaves $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$, where $f_*\mathcal{O}_X$ is the pushforward of $\mathcal{O}_Y$ by f .
- The morphism of sheaves $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is a local homomorphism, i.e.

$$(f^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$$

where $\mathfrak{m}_x \overset{\max}{\trianglelefteq} \mathcal{O}_{X,x}$ and $\mathfrak{m}_{f(x)} \overset{\max}{\trianglelefteq} \mathcal{O}_{Y,f(x)}$.

Remark **Intuition** **Change**: the (3) into the form of $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$. **should be** $\mathcal{O}_{X,x}$ instead of $(f^*\mathcal{O}_Y)_{f(x)}$.

The condition $(f^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$ can be reformulated as $f^\#(\mathfrak{m}_{f(x)}) \subseteq \mathfrak{m}_x$:

Because $(X, \tau, \mathcal{O}_X)$ is a locally ringed space, $\mathcal{O}_{X,x}$ is a local ring. Let its unique maximal ideal be $\mathfrak{m}_x$.

$\mathfrak{m}_x$ is prime. So, $(f^\#)^{-1}(\mathfrak{m}_x)$ is a prime ideal of $\mathcal{O}_{Y,f(x)}$.

$(Y, \sigma, \mathcal{O}_Y)$ is a locally ringed space as well. So, $\mathcal{O}_{Y,f(x)}$ is a local ring, unique maximal ideal $\mathfrak{m}_{f(x)}$. By uniqueness and maximality of $\mathfrak{m}_x$, we know that

$$(f^\#)^{-1}(\mathfrak{m}_x) \subseteq \mathfrak{m}_{f(x)}$$

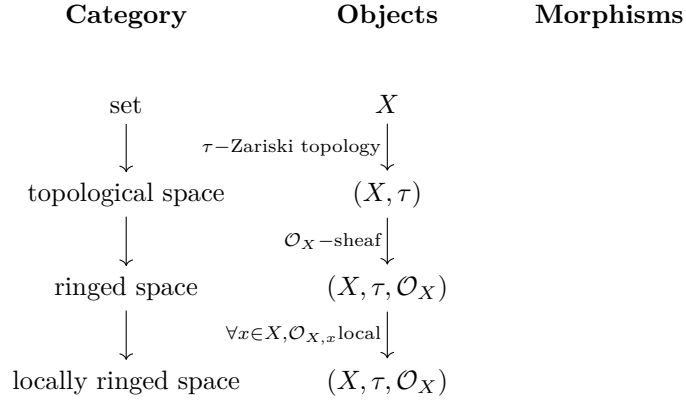
So, to say $(f^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$, it suffices to recast it as $\mathfrak{m}_{f(x)} \subseteq (f^\#)^{-1}(\mathfrak{m}_x) \xleftrightarrow{\dagger} f^\#(\mathfrak{m}_{f(x)}) \subseteq \mathfrak{m}_x$.

$\dagger$: $\Rightarrow$ Assume $\mathfrak{m}_{f(x)} \subseteq (f^\#)^{-1}(\mathfrak{m}_x)$, then $f^\#(\mathfrak{m}_{f(x)}) \subseteq f^\# \circ (f^\#)^{-1}(\mathfrak{m}_x) \subseteq \mathfrak{m}_x$.

$\Leftarrow$ Assume $f^\#(\mathfrak{m}_{f(x)}) \subseteq \mathfrak{m}_x$, then $\mathfrak{m}_{f(x)} \subseteq (f^\#)^{-1} \circ f^\#(\mathfrak{m}_{f(x)}) \subseteq (f^\#)^{-1}(\mathfrak{m}_x)$.

Example **must have**

Here is a diagram to upshot the hierarchy of different concepts.



Proposition 2.1: Global-Local

Let $\mathcal{F}, \mathcal{G}$ be sheaves of rings on (X, τ) . If $\exists$ a morphism of sheaves $\eta : \mathcal{F} \rightarrow \mathcal{G}$, then η induces a ring homomorphism η_x between rings $\mathcal{F}_x$ and $\mathcal{G}_x$:

$$\eta_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x$$

where $\mathcal{F}_x := \varinjlim_{U(\ni x) \in \tau} \mathcal{F}(U)$ and $\mathcal{G}_x := \varinjlim_{U(\ni x) \in \tau} \mathcal{G}(U)$

Proof: ¹ This is directly from morphism between direct system induces a morphism of direct limits. □

Proposition 2.2: Local-Global

Given a morphism between two ringed spaces $(f, f^\#) : (X, \tau, \mathcal{O}_X) \rightarrow (Y, \sigma, \mathcal{O}_Y)$.

If $\forall x \in X$, the induced stalk homomorphism $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ is an isomorphism, then $f^\#$ is an isomorphism of the corresponding structure sheaves.

2.2.1 Sheafification

Definition 2.6: Sheafification

Let $\mathcal{F}$ be a presheaf. The **sheafification** of $\mathcal{F}$ is a sheaf $\mathcal{F}^{\text{sh}}$ such that

$\exists$ morphism of presheaf $\alpha : \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$, $\forall$ sheaf $\mathcal{G}$ with a morphism of presheaf $\beta : \mathcal{F} \rightarrow \mathcal{G}$, $\exists!$ morphism of sheaf $\Phi : \mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$ such that the following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{F} & \xrightarrow{\alpha\text{-pre}} & \mathcal{F}^{\text{sh}} \\
 & \searrow \beta\text{-pre} & \downarrow \exists! \Phi\text{-sheaf} \\
 & & \mathcal{G}
 \end{array}$$

¹Note that we don't know if $(X, \tau, \mathcal{F})$ ($\mathcal{G}$ and so on) is a locally ringed space, so we don't know in priori if $\mathcal{F}_x$ and $\mathcal{G}_x$ are local rings.

2.3 Local properties of sheaves

3 Schemes

3.1 Preliminaries

Proposition 3.1: $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ a ringed space

Let R be a commutative ring and let $\text{Spec}(R)$ be its spectrum. Define $\mathcal{O}_{\text{Spec}(R)}$ as the following: $\forall U \overset{\text{open}}{\subseteq} \text{Spec}(R)$, let

$$\mathcal{O}_{\text{Spec}(R)}(U) := \{f | f \text{ is a function on } U \text{ such that } \forall [\mathfrak{p}] \in U, \exists V \overset{\text{open}}{\subseteq} U, f([\mathfrak{p}]) \in R_{\mathfrak{p}}\}$$

Then, $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is a ringed space.

Proposition 3.2: Stalk structure of $\mathcal{O}_{\text{Spec}(R)}$

$$\mathcal{O}_{\text{Spec}(R), [\mathfrak{p}]} \cong_{\mathbf{CRing}} R_{\mathfrak{p}}$$

3.2 Affine schemes and schemes

3.2.1 Affine scheme

Definition 3.1: Affine scheme

Let $(X, \tau, \mathcal{O}_X)$ be a locally ringed space. $(X, \tau, \mathcal{O}_X)$ is an **affine scheme** $\Leftrightarrow \exists R \in \mathbf{CRing}$,

$$(X, \tau, \mathcal{O}_X) \cong (\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$$

where $\cong$ is an isomorphism between **locally** ringed spaces.

Example (1) $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ itself is an affine scheme.

3.2.2 Scheme

Definition 3.2: Scheme

A scheme $(X, \mathcal{O}_X)$ is a locally ringed space that admits an open cover $\{U_i\}_i$ of X , such that $\forall i, (U_i, \mathcal{O}_X|_{U_i})$ is an affine scheme.

3.2.3 Zariski Tangent space

This is an example to see how the locally ringed space 2.4 is applied. Let's consider the simplest case: The stalk of $\mathcal{O}_X$ at $x \in X$ is explicitly considered as the collections of germs of functions $\text{Germ}_x(X)$. So, for every open neighbourhood U of x , we have a function

$$\psi_U : \mathcal{O}_x(U) \rightarrow \mathcal{O}_{X,x} := \text{Germ}_x(X) \quad f \mapsto f_x$$

as suggested above. At the same time, $\mathcal{O}_{X,x}$ is a local ring with the unique maximal ideal $\mathfrak{m}_x$. Let f_x be an element of $\mathcal{O}_{X,x}$. From a proposition in local rings, $f_x \notin \mathcal{O}_{X,x}^\times \Leftrightarrow f_x \in \mathfrak{m}_x \Leftrightarrow f(x) := \overline{f_x} = 0$ in $\kappa(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$. Let $\lambda := f_x - g_x$ be the constant, then $f_x = \lambda + g_x$. We know how to differentiate a constant λ . Thus, the differentiation of any f at x , $D : \mathcal{O}_{X,x} \rightarrow \kappa(x)$ can be reduced to its maximal ideals $g_x \in \mathfrak{m}_x$. So, D can be streamlined to $D : \mathfrak{m}_x \rightarrow \kappa(x)$. Furthermore, for two functions $g_x, h_x \in \mathfrak{m}_x$, the differentiation of their product should be $D(f_x g_x) = D(f_x)g_x + f_x D(g_x) = \mathfrak{m}_x = \bar{0}$ via the Leibniz rule. So, product of $\mathfrak{m}_x, \mathfrak{m}_x^2$, lies in the kernel of D . The new map $D : \mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow \kappa(x)$ should be well-defined.

² $\mathcal{O}_{X,x}^\times$: units of $\mathcal{O}_{X,x}$.

3.2.4 Tangent space

This is a local construction. To define the tangent space at a point $x \in X$, we focus on the local behavior of functions. The natural domain for differentiation is the stalk $\mathcal{O}_{X,x}$, which captures the germs of functions near x .

A tangent vector at x is defined as a k -linear derivation $D : \mathcal{O}_{X,x} \rightarrow \kappa(x)$, satisfying the Leibniz rule:

$$D(f_x g_x) = f(x)D(g_x) + g(x)D(f_x)$$

where $f(x), g(x) \in \kappa(x)$ are the values of the germs at x .

We can simplify this structure by analyzing the maximal ideal $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$:

1. **Decomposition:** Any germ $f_x \in \mathcal{O}_{X,x}$ can be uniquely written as $f_x = \lambda + g_x$, where $\lambda = f(x)$ is a constant (identifying $k \subset \mathcal{O}_{X,x}$) and $g_x \in \mathfrak{m}_x$ is a germ vanishing at x .
2. **Vanishing on Constants:** Since D is k -linear and satisfies the Leibniz rule, $D(\lambda) = 0$ for any constant $\lambda \in k$. Thus, $D(f_x) = D(g_x)$. The derivation is completely determined by its action on $\mathfrak{m}_x$.
3. **Vanishing on $\mathfrak{m}_x^2$:** Consider any two elements $g_x, h_x \in \mathfrak{m}_x$. Their product $g_x h_x$ lies in $\mathfrak{m}_x^2$. Applying the Leibniz rule:

$$D(g_x h_x) = g(x)D(h_x) + h(x)D(g_x) = 0 \cdot D(h_x) + 0 \cdot D(g_x) = 0$$

Here we used the fact that $g(x) = 0$ and $h(x) = 0$.

Conclusion: Any derivation D vanishes on $\mathfrak{m}_x^2$. Consequently, D induces a well-defined linear map on the quotient space:

$$\bar{D} : \mathfrak{m}_x / \mathfrak{m}_x^2 \rightarrow \kappa(x)$$

This creates a natural isomorphism between the tangent space $T_x X$ (the space of derivations) and the dual of the cotangent space $(\mathfrak{m}_x / \mathfrak{m}_x^2)^*$.

3.2.5 Projective scheme

Let S be a graded ring. So, naturally $S = \bigoplus_{d \geq 0} S_d$. The irrelevant ideal $S_+ = \bigoplus_{d > 0} S_d$, which is a homogeneous ideal. A homogeneous ideal of S is written as $\mathfrak{p}_h$ to distinguish it from an arbitrary ideal $\mathfrak{p}$.

$\text{Proj } S$ is defined to be the collection of all homogeneous prime ideals of S that **do not contain all of** S_+ :

$$\text{Proj } S := \{[\mathfrak{p}_h] : \mathfrak{p}_h \leq S \text{ is prime, } S_+ \not\subseteq \mathfrak{p}_h\}$$

Closed sets of $\text{Proj } S$ are defined to be $V(\mathfrak{p}_h) := \{[\mathfrak{q}] \in \text{Proj } S : \mathfrak{p}_h \subseteq \mathfrak{q}\}$.

The following lemma checks the collection of $V(\mathfrak{p}_h)$ indeed play the role of closed sets.

Lemma 3.1: Closed sets of $\text{Proj } S$

Closed

Before we define the concept of sheaf, let's define the ring $S_{(\mathfrak{p}_h)}$. $S_{(\mathfrak{p}_h)}$ is defined to be the homogeneous elements of degree 0 of the localization $T^{-1}S$ with T the multiplicative set $\bigcup_d S_d \setminus \mathfrak{p}_h$.

Explicitly,

$$S_{(\mathfrak{p}_h)} = \left\{ \frac{f}{g} : \exists d > 0, f, g \in S_d, g \notin \mathfrak{p}_h \right\} = \left\{ \frac{f}{g} : \deg(f) = \deg(g) > 0, f, g \text{ are homogeneous, } g \notin \mathfrak{p}_h \right\}$$

So, $S_{(\mathfrak{p}_h)}$ is a subring of the localized ring $S_{\mathfrak{p}_h}$ consisting of the elements of degree 0.

Now, we try to define the structure sheaf of $\text{Proj } S$, denoted $\mathcal{O}_{\text{Proj } S}$, as follows: For an open subset U of $\text{Proj } S$, define

$$\mathcal{O}_{\text{Proj } S}(U) = \left\{ s : U \rightarrow \prod_{\mathfrak{p}_h} S_{(\mathfrak{p}_h)} : s \text{ is locally quotient} \right\}$$

More explicitly,

$$\mathcal{O}_{\text{Proj } S}(U) = \left\{ s : U \rightarrow \prod_{\mathfrak{p}_h} S_{(\mathfrak{p}_h)} : \forall \mathfrak{p}_h \in U, \exists V(\ni \mathfrak{p}_h) \subseteq U \text{ open, } \forall \mathfrak{q}_h \in V, s(\mathfrak{q}_h) = \frac{a}{f} \in S_{(\mathfrak{p}_h)} \right\}$$

$\mathcal{O}_{\text{Proj } S}$ is a sheaf: (1) (2)

(3) Let $\{U_i\}_i$ be an open cover of U . $\forall i, s|_{U_i} = 0$. $\forall \mathfrak{p}_h \in U$, $\exists i, \mathfrak{p}_h \in U_i$, $s(\mathfrak{p}_h) = s|_{U_i}(\mathfrak{p}_h) = 0$.

(4) Still, take the same $\{U_i\}_i$. Define $s : U \rightarrow \prod_{\mathfrak{p}_h} S_{(\mathfrak{p}_h)}$ by $\mathfrak{p}_h \mapsto s_i(\mathfrak{p}_h)$ where i is the index of U_i that $\mathfrak{p}_h \in U_i$. This definition is well-defined because for i, j such that $\mathfrak{p}_h \in U_i \cap U_j$, $s_i(\mathfrak{p}_h) = s_i|_{U_i \cap U_j}(\mathfrak{p}_h) = s_j|_{U_i \cap U_j}(\mathfrak{p}_h) = s_j(\mathfrak{p}_h)$. Let's define a basis of open sets. $\forall f \in S_+$ homogeneous, i.e. $\forall f \in \bigcup_{d>0} S_d$,

$$D_+(f) := \{[\mathfrak{p}_h] \in \text{Proj } S : f \notin \mathfrak{p}_h\}$$

$D_+(f)$ is open because $D_+(f) = \text{Proj } S \setminus \{[\mathfrak{p}_h] \in \text{Proj } S : f \in \mathfrak{p}_h\} = \text{Proj } S \setminus V(\langle f \rangle)$. $\langle f \rangle$ is generated by a homogeneous element, so it is a homogeneous ideal that does not contain S_+ and $V(\langle f \rangle)$ is a closed set in S .

Proposition 3.3: Relative structures of $(\text{Proj } S, \mathcal{O}_{\text{Proj } S})$

Let S be a graded ring.

(1) Stalk of $\mathcal{O}_{\text{Proj } S}$: the structure of the stalk of $\mathcal{O}_{\text{Proj } S}$ at $[\mathfrak{p}_h] \in \text{Proj } S$, $\mathcal{O}_{\text{Proj } S, \mathfrak{p}_h}$ is a local ring $S_{\mathfrak{p}_h}$, i.e.

$$\mathcal{O}_{\text{Proj } S, \mathfrak{p}_h} \cong S_{\mathfrak{p}_h}$$

(2) Open sets of $\mathcal{O}_{\text{Proj } S}$: $\{D_+(f)\}_{f \in \bigcup_{d>0} S_d}$ is an open cover of $\text{Proj } S$. In addition, there is an isomorphism of locally ringed space:

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \cong (\text{Spec } S_{(\langle f \rangle)}, \mathcal{O}_{\text{Spec } S_{(\langle f \rangle)}})$$

$S_{(\langle f \rangle)}$ is the subring of S_f of elements of degree 0.

(3) Scheme: $(\text{Proj } S, \mathcal{O}_{\text{Proj } S})$ is a scheme.

Remark (1) just says that $(\text{Proj } S, \mathcal{O}_{\text{Proj } S})$ is a locally ringed space. (2) reveals that this locally ringed space is covered by open affine schemes. Hence, it is a scheme. So, (3) is a consequence of (2). □

Proof:

Remark Let R be a ring. The projective n -space over R is a scheme $\mathbb{P}^n(R) \cong \text{Proj } R[x_1, \dots, x_n]$ Up to now, we have more correspondence between the language from classic AG and the schemes.

	affine cases	projective cases
dd	$\text{Spec } R[x_1, \dots, x_n] \cong \mathbb{A}^n(R)$	$\text{Proj } R[x_1, \dots, x_n] \cong \mathbb{P}^n(R)$

3.2.6 Product of schemes

Definition 3.3: Schemes over a fixed scheme

Let $(S, \mathcal{O}_S)$ be a fixed scheme. A scheme $(Y, \mathcal{O}_Y)$ is **over** $(S, \mathcal{O}_S) \Leftrightarrow$ there is a morphism of schemes $(\varphi, \varphi^\#) : (Y, \mathcal{O}_Y) \rightarrow (S, \mathcal{O}_S)$.

A morphism between two schemes $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ over $(S, \mathcal{O}_S)$ is a morphism of schemes $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ that commutes with the morphism to $(S, \mathcal{O}_S)$, i.e. a morphism $(f, f^\#)$ that makes the following diagram commute:

$$\begin{array}{ccc} (X, \mathcal{O}_X) & \xrightarrow{(f, f^\#)} & (Y, \mathcal{O}_Y) \\ & \searrow (\psi, \psi^\#) \quad \swarrow (\varphi, \varphi^\#) & \\ & (S, \mathcal{O}_S) & \end{array}$$

3.3 Correspondence between variety language and scheme language

3.4 Properties of schemes

3.4.1 Local properties: Reducedness, Normality, Regularity and singularity, Cohen-Macaulay

Definition 3.4: Reducedness

A scheme $(X, \mathcal{O}_X)$ is reduced $\Leftrightarrow \forall$ open set of X , the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

3.4.2 Topological properties: Connectedness, Irreducibility, Quasi-compactness

A scheme $(X, \mathcal{O}_X)$ is connected $\Leftrightarrow$ its topological space X is connected.

3.4.3 Mixed: Integral

3.4.4 Gluing properties: Separatedness, properness

In the following sections, we attempt to translate the properties of classical topology into the language of schemes. While these properties (such as Hausdorff and compactness) superficially appear to be ontological attributes of an "absolute space", in a deeper categorical sense, they are all relative properties of morphisms. The illusion that they are spatial properties simply stems from the fact that classical geometry implicitly binds these spaces to a morphism targeting a specific terminal object.

The ultimate goal of this cross-universe translation is to replicate "perfect spaces" from classical geometry within the scheme universe. In layman's terms, it is to preserve locally compact Hausdorff spaces, as these well-behaved spaces possess irreplaceable value in analysis and geometry.

Separatedness

In classical topology, the Hausdorff property is viewed as a pure spatial property. When we say a space X is Hausdorff, our perspective is absolute and introspective—we focus on whether points inside the space can be isolated by disjoint open sets. However, when we attempt to reproduce this property in the universe of schemes, it completely fails. First, the underlying Zariski topology of a scheme is extremely coarse; open sets are very large and inevitably intersect, making it impossible to find disjoint open sets. Second, the underlying topological space of a scheme does not accurately reflect all of its properties. This is not just because the Zariski topology is too coarse, but also because it is highly "distorted". The underlying topological space is completely blind to rich algebraic structures (such as a "fat point" with a first-order infinitesimal perturbation).

Since this topological space (ontology) can neither separate points nor perceive algebraic details, it completely loses its qualification as a concept carrier. Since finding open sets is impossible, we return to the classical universe of topological spaces. There is a characterization of Hausdorff: a classical topological space X is Hausdorff if and only if its diagonal Δ is a closed set in the product space $X \times X$. This characterization completely abandons "open sets within the space" and instead uses an external mapping (the diagonal). Grothendieck astutely grasped this equivalent. In the scheme universe, he did not demand disjoint open sets in the underlying space; rather, he directly used this diagonal morphism to enforce the following definition: as long as the diagonal morphism $\Delta : X \rightarrow X \times X$ of a scheme X is a closed immersion, it is forcibly defined to possess a property analogous to Hausdorff, which is named *separatedness*. Through this specific operation, schemes successfully inherit the geometric soul of unique limits without the Hausdorff topological shell.

From the specific operation of transforming Hausdorff into separatedness described above, we can extract a methodology/philosophy:

(i) Finding higher-dimensional projections: In the classical geometric universe (ontological space), seek the equivalent of the property (e.g., a space being Hausdorff is equivalent to its diagonal being a closed set in the product space).

(ii) Extracting relational essence: Through this equivalent, transform the perspective that originally focused on the internal composition of the space into a purely external relational perspective (i.e., morphisms).

(iii) Cross-universe reconstruction: Project this "pure relationship" stripped of its entity into the scheme universe. As long as the algebraic condition of this external mapping is met (i.e., the diagonal morphism Δ is a closed immersion), we forcibly define it to possess this property.

The diagram below illustrates the series of philosophical processes discussed above:

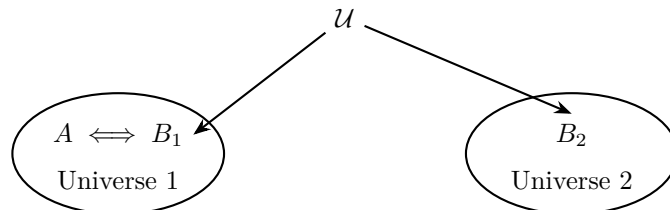


Figure 1: $\mathcal{U}$ and its projections to different universes

In this diagram 1, the symbols connect to our specific context as follows:

A represents the Hausdorffness defined using disjoint open sets in classical topology.

B_1 represents the equivalent condition in classical topological spaces where the diagonal is a closed set.

$\mathcal{U}$ represents the most general relational conclusion.

B_2 represents the ultimate definition we establish in the universe of schemes.

Once this mapping relationship is established in the new universe, we reproduce that geometric soul within the new scheme universe. This methodology also explains why the direct translation of concepts fails:

When we forcefully translate the Hausdorff property from classical geometry into the Zariski topological universe of schemes, it inevitably faces complete failure. But this is not because our descriptive languages are restricted in the two spaces, but because we chose the wrong object to transport. In classical geometry, we attempt to define Hausdorff by finding two disjoint open sets to isolate points. However, this approach merely captures an ontological manifestation of the higher-dimensional relational essence (uniqueness of limits) projected onto the medium of classical topology. If we rigidly transport that ontological manifestation attached to the old medium (namely, the open-set description of Hausdorffness), this non-essential feature will inevitably be severely distorted by the shift in the universe's laws, ultimately leading to theoretical collapse.

Returning to the initial definition of separatedness via the diagonal morphism. In this definition, a turning point is hidden within the product operation $X \times X$. By deeply examining the definition of the product, we find that this so-called 'absolute' product actually implies a base, namely the terminal object $S = \text{Spec}(\mathbb{Z})$ in the category. This means that $X \times X$ is essentially a fiber product over the terminal object in mathematics:

$$X \times X \cong X \times_{\text{Spec}(\mathbb{Z})} X$$

Once we make this implicit terminal object explicit, the perspective changes: the fiber product $X \times_S X$ depends on the structure morphism $X \rightarrow S$ from X to the base S . Thus, the separatedness of the space X we initially discussed is actually not defined on an absolute space. As long as we use the fiber product, we are already discussing the separatedness of a mapping (morphism). Therefore, the so-called 'spatial property' is merely a certain mapping characteristic of the space relative to the base S . Since this property is logically bound to the morphism $X \rightarrow S$, we ought to generalize it to a broader context: we can directly extend this separatedness relative to the base S to a mapping between any two schemes X and Y . During this generalization, we only need to adjust the base of the fiber product, switching it from the implicit terminal object S to the target scheme Y . From this, we arrive at the general definition:

Definition 3.5: Separatedness

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two schemes. Let $(f, f^\#) : X \rightarrow Y$ be a morphism between these two schemes. $(f, f^\#)$ is said to be **separated** $\Leftrightarrow$ the diagonal $\Delta : X \rightarrow X \times_Y X$ is a closed immersion.

Remark The shift in concept binding (from space to morphism) is rooted in algebraic geometry's complete deconstruction of "absolute space". In Grothendieck's geometric universe, absolutely isolated spaces do not exist. The significance of every scheme's existence is contained within its mapping relationships with other schemes. Shifting from "properties of spaces" to "properties of morphisms" is not a deliberate complication, but a return to the essence. It marks the leap of geometry from "ontology" to "relationalism": all geometric truths will ultimately manifest as characteristics of morphisms.

Universally closed

Regarding compactness, we must make a slight terminological distinction: we reserve the word compactness for spaces that are simultaneously quasi-compact and Hausdorff. Quasi-compactness itself merely indicates that every open cover has a finite subcover. This description still operates entirely within the internal ontological perspective of a space. Mimicking our previous logic for reconstructing the Hausdorff property, we must now completely abandon internal open sets and seek an external morphic equivalent. Remarkably, this external mapping exists. We must realize that when discussing the quasi-compactness of a space X , we are actually discussing the properties of a structure morphism $X \rightarrow \{*\}$, where $\{*\}$ is the terminal object in the category of topological spaces (**Top**). In this way, the quasi-compactness of a space (morphism) is perfectly linked to the universal-closedness of the morphism $X \rightarrow \{*\}$. The most general version of $X \rightarrow \{*\}$ is a map $f : X \rightarrow Y$ between two topological spaces. The definition should be:

$X \rightarrow Y$ (continuous) is universally closed $\Leftrightarrow$ for any test space T , $X \times T \rightarrow Y \times T$ is a closed map.

When the target space is $Y = \{*\}$, this equivalence dictates that for any topological space T , the projection $X \times T \rightarrow \{*\} \times T \cong T$ is a closed map. This happens to be precisely the classical equivalent formulation for the

quasi-compactness of space X . Therefore, the quasi-compactness of an absolute space is perfectly externalized as the universal closedness of the morphism $X \rightarrow \{*\}$. This universal closedness is exactly the relational essence that can be losslessly transported to the universe of schemes.

Mapping this to our previous diagram 1, the corresponding entities are:

A : Quasi-compactness (internal ontological manifestation based on open covers).

B_1 : Universal closedness of the morphism $X \rightarrow \{*\}$ (the extracted relational essence).

$\mathcal{U}$: The generalized concept of universal closedness across universes.

B_2 : Universally closed morphisms in the scheme universe (the core pillar of Properness).

However, in the description of universal closedness for general topological spaces (i.e., " $\forall$ topological space T , $X \times T \rightarrow Y \times T$ is a closed map"), the absolute "topological space T " still remains. This implies we must also upgrade the test space T from an absolute object to a relative morphism. How? Just as we replaced X with $X \rightarrow \{*\}$, we replace T with $T \rightarrow \{*\}$. Consequently, absolute products manifest as fiber products: $X \times T \cong X \times_{\{*\}} T$, and $Y \times T \cong Y \times_{\{*\}} T$. To pursue ultimate abstraction, replace again the test map $T \rightarrow \{*\}$ with a continuous map $T \rightarrow Z$ over an arbitrary base space Z , the absolute products should be upgraded to $X \times_Z T$ and $Y \times_Z T$. We might seemingly write the most general definition:

$X \rightarrow Y$ is universally closed $\Leftrightarrow \forall$ continuous map $T \rightarrow Z$, $X \times_Z T \rightarrow Y \times_Z T$ is a closed map.

The maps $X \rightarrow Y$, $T \rightarrow Z$ and $X \times_Z T \rightarrow Y \times_Z T$ can be reconciled into one diagram:

$$\begin{array}{ccccc}
 & & X \times_Z T & & \\
 & \swarrow & \downarrow & \searrow & \\
 T & \xleftarrow{\quad} & & \xrightarrow{\quad} & Y \times_Z T \\
 \downarrow & & \downarrow & & \downarrow \\
 & & X & & \\
 \downarrow & \swarrow & & \searrow & \downarrow \\
 Z & \xleftarrow{\quad} & & \xrightarrow{\quad} & Y
 \end{array}$$

We can test this grand unified formula using our previous equivalent formulation. When the base $Z = Y$, $X \rightarrow Y$ being universally closed means that $\forall$ continuous map $T \rightarrow Y$, the map $X \times_Y T \rightarrow Y \times_Y T$ should be closed. Since $Y \times_Y T \cong T$, the right side perfectly collapses into T . But if we forcefully test the absolute map $X \rightarrow \{*\}$ using an arbitrary base Z , the right side becomes $\{*\} \times_Z T$, which is not isomorphic to T . This demonstrates that relaxing the condition to "for any continuous map $T \rightarrow Z$ " leads to a categorical type mismatch (it is too broad). To evaluate the morphism $X \rightarrow Y$, the base of the pullback must strictly match its target space, thus it must be corrected to $T \rightarrow Y$ ³. From this, we finally establish the most general and rigorous definition of a universally closed mapping in the scheme universe:

A morphism $X \rightarrow Y$ is universally closed $\Leftrightarrow \forall$ continuous map (base change) $T \rightarrow Y$, the pullback projection $X \times_Y T \rightarrow Y \times_Y T \cong T$ is a closed map.

³Also hinted by the equivalence when Y is Hausdorff.

4 Sheaf of modules

4.1 $\mathcal{O}_X$ -modules, ideal sheaves

Definition 4.1: $\mathcal{O}_X$ -modules

Let $(X, \tau, \mathcal{O}_X)$ be a ringed space. A **sheaf of $\mathcal{O}_X$ -modules** or **$\mathcal{O}_X$ -module** over $(X, \tau, \mathcal{O}_X)$ is a sheaf $\mathcal{F}$ such that

(1) $\forall U \subseteq^{\text{open}} X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module.

(2) For any sheaf satisfies (1), $\mathcal{F}$, it is compatible with the restriction map. i.e. $\forall V \subseteq U$ and $V, U \in \tau$, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}_X(U) \times \mathcal{F}(U) & \xrightarrow{\text{action}} & \mathcal{F}(U) \\ \text{res}'_{U,V} \times \text{res}_{U,V} \downarrow & & \downarrow \text{res}_{U,V} \\ \mathcal{O}_X(V) \times \mathcal{F}(V) & \xrightarrow{\text{action}} & \mathcal{F}(V) \end{array}$$

Remark To further decode this, using element $r \in \mathcal{O}_X(U)$ and $f \in \mathcal{F}(U)$, through one path we finally have $r|_V \cdot f|_V$ and through another path we have $r \cdot f|_V$. Compatibility $\Leftrightarrow r|_V \cdot f|_V = r \cdot f|_V$.

4.2 Quasi-coherent sheaves: 'Finite dimensional' modules

4.3 Coherent sheaves: The 'Noetherian' case

5 Projective schemes and projective morphisms: Example of a quasi-coherent sheaves

5.1 Quasi-coherent sheaf on $\text{Proj } S$

5.2 Serre's twisting sheaves $\mathcal{O}(n)$

5.3 Blow-ups

6 Sheaf cohomology: machinery

measure of failure of gluing local information into global

6.1 Some motivations

6.2 Čech cohomology

Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open cover of X . The idea behind the Čech cohomology is looking at the further and further intersections:

Let's first talk about the complex or the elements we are dealing with:

(1.1) We look at the intersections $U_i \cap U_j$ between any two covers U_i and U_j . On each U_i , there is an $s_i \in \Gamma(U_i, \mathcal{F})$. The goal is to compare if $s_i = s_j$ on their intersections $U_i \cap U_j$. More precisely, whether

$$U_i|_{U_i \cap U_j} = U_j|_{U_i \cap U_j} \text{ or } s_i|_{U_j} = s_j|_{U_i}$$

⁴ Since this process is ranging over all the U_i s (all $\Gamma(U_i, \mathcal{F})$), we should pack those abelian groups up. Hence, we consider $\prod_i \Gamma(U_i, \mathcal{F})$ in this step. This product is the first complex and we can assign it a symbol $C^0(\mathfrak{U}, \mathcal{F})$, namely

$$C^0(\mathfrak{U}, \mathcal{F}) := \prod_i \Gamma(U_i, \mathcal{F})$$

⁴The second one is because s_i is already defined on U_i so if we just restrict it to U_j , then it is already restricted to $U_i \cap U_j$.

(1.2) We further examine the elements on the two intersections. Because we still want to consider all the two intersections, represented as $\prod_{i_0 < i_1} \Gamma(U_{i_0} \cap U_{i_1}, \mathcal{F})$. This is the second complex, denoted

$$C^1(\mathfrak{U}, \mathcal{F}) = \prod_{i_0 < i_1} \Gamma(U_{i_0} \cap U_{i_1}, \mathcal{F})$$

To make it effortless in generalization of the notations, we adopt the following notion from now on:

$$U_{i_0 i_1 \dots i_n} = \bigcap_{i_0 < i_1 < \dots < i_j < \dots < i_n} U_{i_j}$$

In general, the q -complex is

$$C^q(\mathfrak{U}, \mathcal{F}) = \prod_{i_0 < i_1 < \dots < i_q} \Gamma(U_{i_0 i_1 \dots i_q}, \mathcal{F})$$

By the axiom of sheaves we know that if there are many globally defined functions on each open set U , then as long as they match on each intersection, we can finally extend this function to a global one. Now we are trying to measure if two functions on each open set are equal on their intersections and measure the topological obstructions. Take the C_1 complex for some motivation. Every element $s \in C^1(\mathfrak{U}, \mathcal{F})$ has ij -components s_{ij} for all i and j with $i < j$. And we can think of this s_{ij} as a 'height difference' function. This 'height difference' function s_{ij} might correspond to some function s_i on U_i and s_j on U_j . More precisely, $s_i \in \Gamma(\mathfrak{U}_i, \mathcal{F})$ and $s_j \in \Gamma(\mathfrak{U}_j, \mathcal{F})$. Normally, this s_i and s_j do not match at their intersection $U_i \cap U_j$. But this might be caused by 'artificial choices'. If $s_{ij} = s_i|_{U_j} - s_j|_{U_i}$, then this means this 'height difference' between s_i and s_j can be fixed by transporting s_i by s_{ij} a little bit. Up to this fixing or this transporting, the two functions defined on U_i and U_j can finally equal. So in this case we admit that this final equality case does not have any topological obstructions to extending the local functions into a global one. This information could be encraved into a boundary operator, δ . For some $h \in C^0(\mathfrak{U}, \mathcal{F})$, This δ is defined componentwise as

$$(\delta h)_{ij} = h_j|_{U_i} - h_i|_{U_j}$$

For this $s \in C^1(\mathfrak{U}, \mathcal{F})$ to be able to fix, we need this ij -component of s , s_{ij} . Each ij component of S is a difference as *one* corresponding component in the $C^0(\mathfrak{U}, \mathcal{F})$. This means $s \in \text{im} \delta$.

In the meanwhile, If s is in the image of δ , then it satisfies a cocycle condition. This cocycle condition is equivalent to saying that $\text{im} \delta \subseteq \ker \delta$ and equivalent to $\delta^2 = 0$.

Here is a diagram showing what are the trivial cases and some non-trivial cases causing the topological abstractions. One can also observe that this δ can be generalized to arbitrary q -complex and the index for each component is the boundary operator to the simplicial set.

6.3 Cohomology of $\mathcal{O}_X$ -modules

7 Sheaf cohomology: exmaples of calculations

7.1 An important example: cohomology of a projective scheme $H^i(\mathbb{P}^n(k), \mathcal{O}(m))$

7.2 Cohomology of quasi-coherent modules

8 Bundles and divisors: Toolkits for geometors

8.1 Line bundles: Locally free sheaves

8.1.1 Vector bundles

Definition 8.1: Real vector bundle

A real vector bundle is a tuple $(X, E, \pi, \{\varphi_U\}_U)$ such that:

- (1) X is a topological space, called the base space.
- (2) E is a locally trivial topological space, called the total space.
- (3) $\pi : E \rightarrow X$ is a continuous surjective map, called the bundle projection.
- (4) $\forall x \in X, \pi^{-1}(\{x\})$ is a finite dimensional vector space.

The local triviality of E means that $\forall$ open set U of X , $\pi^{-1}(U)$ is homeomorphic to $U \times \mathbb{R}^k$, i.e. $\exists$ a homeomorphism $\varphi_U : U \times \mathbb{R}^k \rightarrow \pi^{-1}(U)$, called **local trivialization** that makes the following diagram commute:

$$\begin{array}{ccc} \pi^{-1}(U) & \xleftarrow{\varphi_U} & U \times \mathbb{R}^k \\ & \searrow \pi \quad \swarrow p & \\ & U & \end{array}$$

where p is given by: $\forall x \in X, \forall v \in \mathbb{R}^k, p(x, v) = x$

The vector space structure is explicitly given by: Fix an $x \in X$. $\mathbb{R}^k \rightarrow \pi^{-1}(\{x\})$ with $v \mapsto \varphi_U(x, v)$ is an linear isomorphism.

Remark In this definition, as a topological space, $\pi^{-1}(\{x\})$ is homeomorphic to the space $\{x\} \times \mathbb{R}^k$. This comes from the homeomorphism $\varphi_U : U \times \mathbb{R}^k \rightarrow \pi^{-1}(U)$. In the meanwhile, $\pi^{-1}(\{x\})$ also has an algebraic structure, which is given explicitly by the linear isomorphism to $\mathbb{R}^k$, or $\{x\} \times \mathbb{R}^k$. The homeomorphism is not enough to determine the linear isomorphism. Consider an example: $f : \mathbb{R} \rightarrow \mathbb{R}$ by $v \mapsto v^3$, which is a homeomorphism but not a linear map.

In one word, structures of $\pi^{-1}(\{x\})$ align perfectly with structures of $\{x\} \times \mathbb{R}^k$ in two levels.

When $k = 1$, the space E or the tuple $(X, E, \pi, \{U_i\}_i)$ is called a **line bundle**.

When $E = X \times \mathbb{R}^k$, it is called a **trivial bundle**.

Sections of a vector bundle:

Transition functions:

There is one subtlety needs to deal with. Since the local trivializations work on each open subset of X in the definition. For the intersection of two open sets, they have to match to each other.

Suppose there are two local trivializations on U_α and U_β , $\varphi_\alpha : U_\alpha \times \mathbb{R}^k \rightarrow \pi^{-1}(U_\alpha)$ and $\varphi_\beta : U_\beta \times \mathbb{R}^k \rightarrow \pi^{-1}(U_\beta)$. Consider there composition on $U_\alpha \cap U_\beta$, $\varphi_\beta^{-1} \circ \varphi_\alpha : (U_\alpha \cap U_\beta) \times \mathbb{R}^k \rightarrow (U_\alpha \cap U_\beta) \times \mathbb{R}^k$, which maps $(x, v) \in (U_\alpha \cap U_\beta) \times \mathbb{R}^k$ to some $(x, u) \in (U_\alpha \cap U_\beta) \times \mathbb{R}^k$. Since $\varphi_\alpha(x, v) \in \pi^{-1}(\{x\})$ and $\varphi_\beta^{-1}|_{\pi^{-1}(\{x\})} : \pi^{-1}(\{x\}) \rightarrow \{x\} \times \mathbb{R}^k$ is a homeomorphism.

Furthermore, $u = T_x(v)$ for some $T_x \in \text{GL}(\mathbb{R}^k)$. Because $\varphi_\alpha|_{\pi^{-1}}$ and $\varphi_\beta|_{\pi^{-1}}$ are linear isomorphisms. Fix an x .

$$\begin{array}{ccccccc} \mathbb{R}^k & \xrightarrow{\sim} & \{x\} \times \mathbb{R}^k & \xrightarrow{\sim} & \pi^{-1}(\{x\}) & \xrightarrow{\sim} & \{x\} \times \mathbb{R}^k & \xrightarrow{\sim} & \mathbb{R}^k \\ & & \searrow & & \swarrow & & \searrow & & \swarrow \\ v & \xrightarrow{\quad} & (x, v) & \xrightarrow{\quad} & \varphi_\alpha(x, v) & \xrightarrow{\quad} & (x, u) & \xrightarrow{\quad} & u \end{array}$$

is a linear isomorphism. Notice that this linear isomorphism depends on x . So, $x \mapsto T_x$ is actually a map, called $g_{\beta\alpha} : U_\alpha \cap U_\beta \rightarrow \text{GL}(\mathbb{R}^k)$.
 So, more explicitly, for an $x \in U_\alpha \cap U_\beta$, $\varphi_\beta^{-1} \circ \varphi_\alpha(x, v) = (x, (g_{\beta\alpha}(x))(v))$.

Category of vector bundles:

Operations of vector bundles:

8.1.2 Invertible sheaves and locally free sheaves

From now on, let's focus on the line bundle cases ($\pi : E \rightarrow X$ has local trivialization $\varphi : U \times \mathbb{R} \rightarrow \pi^{-1}(U)$). One can encapsulate the sections $s : U \rightarrow E$ into a sheaf as the classic theory does. In the structure sheaf part, a sheaf $\mathcal{O}$ is 'evaluated' on each open set U of X to get $\mathcal{O}(U) = \Gamma(U, \mathcal{O})$. Here, the process is reversed. We first care about all sections and then make them into a sheaf.

Let $\mathcal{L}$ be the 'sheaf of line bundles'. $\mathcal{L}$ is formally defined as follows. For every open set U of X , $\mathcal{L}(U)$ is defined to be the collection of all sections on U of the line bundle E . Namely,

$$\mathcal{L}(U) := \{s : U \rightarrow E \mid \pi \circ s = \text{id}_U\}$$

There is another way to rephrase the element of $\mathcal{L}(U)$ as functions $f_i : U_i \rightarrow \mathbb{R}$ with U_i being open in U . So, there is a correspondence

$$\{\text{sections of } U\} \xrightleftharpoons[\Psi]{\Phi} \{\text{functions } f_i : U_i \rightarrow \mathbb{R} \mid \forall i, j, f_i = g_{ij} \cdot f_j\}$$

where g_{ij} has the same meaning as above-mentioned $g_{\beta\alpha}$. There is a subtlety on the $\cdot$. Formally, $g_{ij} : U \rightarrow \text{GL}(\mathbb{R})$. $(g_{ij} \cdot f_j)(x) = g_{ij}(x) \cdot f_j(x)$. But, since $\text{GL}(\mathbb{R}) \cong \mathbb{R}^\times$, there is no problem to write the operation between g_{ij} and f_j as multiplication. However, for the case $k > 1$, it is not multiplication anymore.

Given a section $s : U \rightarrow \pi^{-1}(U)$, one can get a function $U \cap U_i \rightarrow \mathbb{R}$ by composing $\tilde{p} \circ \varphi^{-1} \circ s$, where $\tilde{p} : U \times \mathbb{R} \rightarrow \mathbb{R}$ is the natural projection of (x, v) to its second coordinate. In a detailed way, $\Phi(s) := \tilde{p} \circ \varphi^{-1} \circ s$ and

$$U \xrightarrow{s} \pi^{-1}(U) \xrightarrow{\varphi^{-1}} U \times \mathbb{R} \xrightarrow{\tilde{p}} \mathbb{R}$$

For a function $f_i : U_i \rightarrow \mathbb{R}$, given $x \in U_i$, a section s_i can be given by $s_i := \varphi_i \circ (\text{id}_{U_i} \times f_i)$. So, we have $\Psi(f_i) := \varphi_i \circ (\text{id}_{U_i} \times f_i)$ and

$$U \cap U_i \xrightarrow{\text{id}_{U_i} \times f_i} (U \cap U_i) \times \mathbb{R} \xrightarrow{\varphi_i} \pi^{-1}(U \cap U_i)$$

This $\Psi(f_i)$ is well-defined. Take any $x \in U \cap U_i \cap U_j$. $\Psi(f_i)(x) = \varphi_i \circ (\text{id}_{U_i} \times f_i)(x) = \varphi_i(x, f_i(x))$. Similarly, $\Psi(f_j)(x) = \varphi_j(x, f_j(x))$. But,

$$\varphi_j(x, f_j(x)) = (\varphi_i \circ \varphi_i^{-1}) \circ \varphi_j(x, f_j(x)) = \varphi_i \circ (\varphi_i^{-1} \circ \varphi_j)(x, f_j(x)) = \varphi_i(x, g_{ij}(x)f_j(x)) = \varphi_i(x, f_i(x))$$

It is quick to see that Ψ is the set-theoretic inverse of Φ .

So, now there are two sets of languages to describe $\mathcal{L}(U)$. Each language has its own nuances. The language of sections is straightforward. The language of

terms	Geometry perspective	Algebraic perspective
	Vector bundle $\pi : E \rightarrow X$	Locally free sheaves $\mathcal{E}$
	Fiber E_x	Sheaf $\mathcal{E} \otimes_{\mathcal{O}_X} \kappa(x)$
	Section $s : X \rightarrow E$	Elements $s \in \Gamma(X, \mathcal{E})$

8.1.3 Correspondence between line bundles and invertible sheaves

8.1.4 Embedding schemes into a projective space

8.1.5 Classification of vector bundles over the projective line

8.2 Divisors

Order of a divisor.

Given a collection of divisors $\{D_1, D_2, \dots, D_n\}$. The maximum is

$$\max_i \{D_1, \dots, D_n\} := \sum_P (\max_i \text{ord}_P(D_i)) [P]$$

where P ranges over all poles and zeros of each D_i .

Once the maximum and minimum is defined, the divisor group can be given a lattice structure. Principal divisors.

8.3 Correspondence between line bundles and divisors

9 Curve Theory

9.1 Riemann-Roch

9.2 Classification of curves in $\mathbb{P}^3$

10 Surface Theory

11 Calculus

11.1 Differentials

11.2 Sheaf of Kähler differentials